SYSDESIGN REFERENCE

Service Discovery Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Let clients find healthy service instances as membership changes.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / SERVICE-DISCOVERY

SYSDESIGN

Service Discovery Tradeoff Matrix

CORE IDEA

Let clients find healthy service instances as membership changes.

Reference architecture decision chain

01 Client-Side Vs Server-Side Discovery	02 Registries	03 Health	04 Ttls

Decisions to defend

+ Client-side vs server-side discovery
+ registries
+ health
+ TTLs
+ and DNS behavior.

Failure modes to test

+ Stale endpoints
+ thundering reconnects
+ and registry outage.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Service Discovery Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

- Partition key
- Versioning
- Uniqueness
- Retention
- Reconciliation

Operations questions

- SLO
- Backlog
- Capacity
- Alert
- Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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